Contents

1 Open and closed sets

Definition. An interval on the number line is called open if it goes from A to B without containing A and B, ie it does not contain its boundary.

The same can be said for subsets of the plane or hyperplanes – they are open if they do not contain any of their boundary. They are closed if they contain all of their boundary. The formal definition of a boundary is a point where no matter how small of a circle or ball you make around that point, it will contain points both inside and outside the set. Note that since an open set contains none of its boundary, it has the property that for any point in the open set, all points sufficiently close to that point are in that set, so open sets must always have some thickness to them.

This is related to the idea that there is no smallest real number greater than another real number, and whether sets are open or closed turns out to be surprisingly important. Also, arbitrary unions of open sets are open since each point has a "ball" around it inside each of the open sets in the union and thus the whole thing, similarly arbitrary intersections of closed sets are closed. The reverse implications only hold for finite unions and intersections (since infinitely many balls intersection could be a point but finitely many is a ball). Sets can be both open and closed, and sets can be neither open nor closed.

A useful way to think of closed sets as sets where the set of everything not in the closed set forms an open set.

We usually call open intervals (a,b) meaning everything from a to b not inclusive, and closed intervals [a,b] to mean everything from a to b inclusive, and [a,b) means everything from a to b including a but not b.

Lemma. Suppose we have a sequence of closed and bounded sets C_n such that for all n we have $C_n \supseteq C_{n+1}$ and they are all non-empty, then their intersection is non-empty.

Proof. We let a_n be the minimum point of C_n, ie its infimum. Then by the fact that this is a nested sequence of sets, a_n is an increasing sequence which is bounded by an upper bound of C_1. Therefore this has a limit which is l. Then I claim that l is in all C_n, which would complete the proof.

To see this, note that if there is some k such that l is not in C_k then by closedness we have that there is an open interval $(l-\varepsilon, l+\varepsilon)$ not in C_k for some $\varepsilon > 0$. Now for this ε let N be large enough such that $a_N > l - \varepsilon$, then the minimum of C_N must be $> l + \varepsilon$ which is a contradiction.

Note that this requires the sets to be closed, since for example the intersection of the open intervals $(0, \frac{1}{2^n})$ is empty, since it does not contain 0 and any number larger than 0 will eventually be removed if n is large enough.

$\square$

2 Cardinality

Now, I will talk about cardinality. This is really fun stuff, but I will not go through all the fun results here, only the results we need. The motivation of this is to show that there are, in a precise sense, "more" real numbers than natural numbers, so I can show that you cannot take a sum over all the real numbers of positive numbers and still get a finite number.

We consider two sets to be the same size if there is a one to one correspondence between their elements. This sounds obvious, but it shows that there are equally as many even positive integers as positive integers, because we can map 1—2 and 2—4 and 3—6 and so on, giving us a one to one correspondence.

Theorem. We cannot find a one to one correspondence between real numbers on any interval and positive integers.

The proof of this theorem is very famous in "popular math" and the slogan is "there are different sizes of infinity".

Proof. We will do this for the interval (0,1) then note that we can get to any interval by scaling, or the entre reals by doing $-\cot(\pi x)$ for example. The property of there being a correspondence to positive integers is called being countable. But, I think listable is a better term.

So, suppose we do, in fact, have a list of all the real numbers between 0 and 1.

1. 0.3141592653589793...
2. 0.2718281828459045...
3. 0.1414213562373095...
4. 0.3498579345858968...
5. 0.9988237478199283...

Now construct a new number where the n'th decimal digit after the point is not the red number, and such that we do not have infinite trailing 9's or 0's (to ensure the number has a unique decimal representation. For the numbers in our list we pick the one with trailing 0's and not trailing 9's whenever we have to make this choice). For example, adding 1 to each red we could have 0.48293... But this is clearly not on the list – It differs from everything on the list by at least one digit. Contradiction. Essentially, this makes it so if you think you found a way to list them, I can prove you are wrong.

$\square$

We want to prove that you can't add an unlistable amount of real numbers greater than 0 and get a finite result.

Now if we had a sum over the real numbers of positive numbers x, split them into subsets:

$$x \geq 1,\ \frac{1}{2} \leq x < 1,\ \frac{1}{3} \leq x < \frac{1}{2},\ \frac{1}{4} \leq x < \frac{1}{3}.$$

Then one of these subsets has to have infinitely many elements. Why? If they all had finitely many elements then they would form a subset of this infinite table:

	1	2	3	4	5	6	7	8	9	...
Set 1										
Set 2										
Set 3										
Set 4										
Set 5										
Set 6										
Set 7										
Set 8										
...										

So I could then go along the table in a zigzag pattern like this in the order shown below, adding to my list whenever the elements are in the table, and not adding them otherwise.

	1	2	3	4	5	6	7	8	9	...
Set 1	1	2	6	7	15	...				
Set 2	3	5	8	14						
Set 3	4	9	13							
Set 4	10	12								
Set 5	11									
Set 6										
Set 7										
Set 8										
...										

All cells get reached eventually, so this would imply we can list the cells. But we are assuming we are adding one term for each real number, and we cannot list the real numbers. Therefore, one of the sets

$$x \geq 1, \ \frac{1}{2} \leq x < 1, \ \frac{1}{3} \leq x < \frac{1}{2}, \ \frac{1}{4} \leq x < \frac{1}{3}.$$

has infinitely many elements, so the sum is infinite.

Note that earlier when I claimed a union of a sequence of measurable sets is measurable, the important thing was that this only allows countable unions. Uncountable sets cannot necessarily be written as the union of a sequence of sets, since a sequence is countable.

3 Properties of integrals

3.1 The extended integral

We've seen the definition of the riemann integral in level 4 - it is the limit of vertical rectangles. We will define the extended integral as follows to help us.

Let I_i be a finite family of n sets which are points, open intervals, or closed intervals, such that they are disjoint (ie the intersection of any 2 is empty). A step function is then any function $s(x) = \sum_{i=1}^{n} c_i \chi_{I_i}(x)$ where χ_{I_i} is the indicator function of I_i, meaning a function that is 1 in I_i and 0 elsewhere.

Define the extended integral of s to be $\sum c_i \text{length}(I_i)$ where length is defined the obvious way. We now need to define this for general functions.

Definition. If there exists a sequence $s_n(x)$ of non-negative step functions that is non-decreasing for every x that approaches f in the limit, then f is integrable from $-\infty$ to ∞ with integral $\lim \int s_n$. By setting f and all s_n to 0 outside some closed interval we get the integral over that finite interval. We will prove shortly that

i) This does not depend on the sequence given such an f

ii) Once we extend this to all real functions f, we will show that this agrees with the Riemann integral when it is defined

Definition. Given sets A, B I will write $A \setminus B$ to mean everything in A but not in B.

Lemma. If a sequence of non-increasing step functions s_n converges to 0 everywhere then $\lim_{n\to\infty}\int s_n = 0$.

Proof. Since s_1 is made of finitely many finite intervals there exist real numbers a, b such that s_1 is 0 outside of $[a, b]$. Also, s_1 is bounded by some maximum value M. Then M is a bound for all s_n since the sequence is decreasing.

For some $\varepsilon > 0$ (the idea being as usual that this can be as small as we want) set $\epsilon = \frac{\varepsilon}{2(b-a)}$. Define $A_n = \{x \in [a, b] \,|s_n(x) \geq \epsilon\}$

Because s_n is a step function this is always just a finite union of intervals, so we can define its length as $l(A_n)$.

Since the functions converge to 0 everywhere each point can only stay in A_n for finitely many steps so the intersection of all of them must be empty.

Note that each A_n may contain open boundaries. We will define K_n as a closed subset of A_n such that the length we removed is $< \frac{\varepsilon}{2^{n+1}M}$ - we will see shortly why we picked those numbers. This is always possible to find since we just remove a tiny part of the end of finitely many intervals if the end is open. For each n we define $F_n = \bigcap_{i=1}^{n} K_i$, so the intersection of the first n of these closed sets. This is now a decreasing sequence of sets, in the sense that $F_n \supseteq F_{n+1}$ always, and $F_n \subseteq A_n$ always.

It is clearly true that the intersection of all F_n is empty since it is contained in the intersection of all A_n. Thus, by the lemma in section 1 of this document there exists some N such that F_N is empty. Therefore, we have

$$A_N = A_N \setminus F_N = A_N \setminus \left(\bigcap_{i=1}^{N} K_i\right) = \bigcup_{i=1}^{N} (A_N \setminus K_i)$$

Since $A_N \subseteq A_i$ for all $i \leq N$ we have

$$A_N \subseteq \bigcup_{i=1}^{N} (A_i \setminus K_i)$$

Now we take the length of both sides:

$$l(A_N) \subseteq \sum_{i=1}^{N} l(A_i \setminus K_i) < \sum_{i=1}^{N} \frac{\varepsilon}{2^{i+1}M} = \frac{\varepsilon}{2M}$$

Now for any $n > N$ we have

$$\int s_n(x)\,dx = \int_{A_n} s_n(x)\,dx + \int_{[a,b]\setminus A_n} s_n(x)\,dx$$

$$< \frac{\varepsilon}{2M} M + \int_{[a,b]\setminus A_n} s_n(x)\,dx = \frac{\varepsilon}{2} + \int_{[a,b]\setminus A_n} s_n(x)\,dx$$

since $l(A_n) < \frac{\varepsilon}{2M}$ and $s_n \leq M$

Now by definition of A_n we have that $s_n < \epsilon = \frac{\varepsilon}{2(b-a)}$ outside of it so we conclude that we get the inequality

$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2(b-a)}(b-a) = \varepsilon$$

Since ε was arbitrary and can therefore be made as small as we want we are done.

□

Proposition. The integral of f does not depend on the sequence f_n

Proof. Suppose that $g_n \to f$ and $f_n \to f$ but their integrals converge to a different value and they are both non-decreasing sequences of functions.

We will prove that $\lim \int g_n \geq \lim \int f_n$. If we can do this then by the same argument $\lim \int f_n \geq \lim \int g_n$ so they are equal.

Fix some integer m, then we know that $f_m \leq f$. For this m define $s_n(x) = \max(0, f_m(x) - g_n(x))$. Then this is a decreasing sequence of step functions that tends to 0 since $g_n \to f$ so the limit of $f_m(x) - g_n(x)$ is ≤ 0.

Thus, $\lim \int s_n = 0$. Since, $f_m \leq g_n + s_n$ for every m, we know that

$$\int f_m \leq \int g_n + \int s_n$$

By the previous lemma $\int s_n \to 0$ so we have

$$\int f_m \leq \lim \int g_n + \lim \int s_n = \lim \int g_n$$

Taking the limit on the left hand side gives

$$\lim \int f_m \leq \lim \int g_n$$

Therefore we are done.

$\square$

Remark. On the above proof we implicitly did something where we ought to be careful. Note that if we have a sequence such that $a_n \leq l$ for all n then we intuitively cannot have $\lim(a_n) > l$. However, be careful, as we can have $a_n \to l$ where $a_n < l$ for all n. Just something to keep in mind.

Theorem. Let f_n be a sequence of non-negative functions integrable in the above sense such that $f_1 \leq f_2 \leq ... \to f$ pointwise, meaning that for each x $f_n(x)$ converges to $f(x)$ as a sequence. Then f is integrable and $\int f = \lim_{n\to\infty} \int f_n$

Note: When working with Riemann integrals and not these extended ones (which often we will be) we do not have that the limit is integrable and so we need to prove it.

Proof. For each f_n let $\phi_{n,m}$ be a non-decreasing sequence of step functions such that as $m \to \infty$, $\phi_{n,m} \to f_n$.

Define

$$\psi_k = \max_{1 \leq n \leq k} \phi_{n,k}$$

First we need to check that this is an increasing sequence of step functions. This is true because if we increase k to $k+1$ then $\phi_{n,k+1} \geq \phi_{n,k}$ so

$$\psi_k = \max_{1 \leq n \leq k} \phi_{n,k} \leq \max_{1 \leq n \leq k} \phi_{n,k+1} \leq \max_{1 \leq n \leq k+1} \phi_{n,k+1} = \psi_{k+1}$$

We now will prove that this is a sequence that converges to f.

Step 1. We will prove that $\lim(\psi_n) \leq f$.

Since the sequence of f_n is increasing, we know that if $n \leq k$ then $\phi_{n,k} \leq f_n \leq f_k$ so it follows that $\psi_k \leq f_k$ for all k. Thus, $\lim(\psi_k) \leq \lim(f_k) = f$

Step 2. We will prove that $\lim(\psi_n) \geq f$.

By the max definition we know that for some n, if $k \geq n$ then $\psi_k \geq \phi_{n,k}$. As $k \to \infty$ the right hand side goes to f_n. Taking limits on both sides gives

$$\psi_k \geq \phi_{n,k}$$

Letting $k \to \infty$ and taking limits we have

$$\lim(\psi_n) \geq f_n$$

Taking limits again gives the desired result.

By steps 1 and 2 we have $\psi_n \to f$ so f is integrable as we have a sequence of step functions.

Step 3. We will show that $\int f = \lim_{n\to\infty} \int f_n$

To do this we will use the inequalities from step 2 and the obvious fact that $a \leq b$ implies $\int a \leq \int b$. Specifically we conclude that

$$\int \psi_k \leq \int f_k \leq \int f$$

Taking limits on both sides and using the definition of the integral, we have

$$\lim \int \psi_k \leq \lim \int f_k \leq \int f$$

$$\int f \leq \lim \int f_k \leq \int f$$

The only way this can happen is that the two things are equal.

□

Theorem. Let f be a riemann integrable function, then its integral in the above sense exists and equals its extended integral

Proof. We will prove this in the subsection on Fubini's theorem shortly

□

We now extend the integral to functions which do not have to be non-negative by saying:

Let $u = \max(0, f)$ and $v = \min(0, f)$, the define $\int f = \int u - \int(-v)$. Basically we do it the stupid way and just split it into positive and negative parts.

As a reminder, liminf of a sequence a_n means the limit as n goes to infinity of $\inf\{a_k : k > n\}$

Lemma. If we have a sequence of non-negative functions $f_n(x)$, then for every x define $f(x) = \text{liminf}_{n\to\infty} f_n(x)$, then on any interval we have $\int f \leq \text{liminf}_{n\to\infty} \int f_n$ provided the integrals exist in the extended sense.

Proof. For all fixed x, we define $g_n(x)$ as $\inf_{k\geq n} f_k(x)$. Then we know that for any fixed x, this is not decreasing as chopping more terms of the start cannot make the infimum lower. We also have the following by stuff we have discussed so far:

We have $\int f = \int \text{sup} g_n = \sup \int g_n$ where the second equality is because g is increasing so we can apply the monotone convergence theorem. Since $\int g_n$ is an increasing sequence, the lim inf of this sequence is going to equal the limit of the sequence, as the infima of the tails are the same as the terms themselves. But the limit of an increasing sequence is the same as its supremum, so we now have that $\int f = \sup \int g_n = \text{liminf}_{n\to\infty} \int g_n$. Because g_n is by definition not greater than f_n for all n at any input value, we have that$\int f = \sup \int g_n = \text{liminf}_{n\to\infty} \int g_n \leq \text{liminf}_{n\to\infty} \int f_n$, completing the proof of Fatou's lemma. From how we defined f, we have that $\int \text{liminf}_{n\to\infty} f_n \leq \text{liminf}_{n\to\infty} \int f_n$.

□

Remark. We can also get that if f_n is bounded above by a function g with a finite integral on our interval for all n, then applying Fatou's lemma to $g - f_n$ gives us the reverse fatou lemma:$\int \text{limsup}_{n\to\infty} f_n \geq \text{limsup}_{n\to\infty} \int f_n$. We will use this to prove the Dominated Convergence Theorem.

3.2 The dominated convergence theorem

This theorem is completely overpowered for making the applied nonsense in A levels/university Maths rigorous.

The **dominated convergence theorem** is as follows:

Theorem. Suppose we have a sequence of integrable functions $f_n(x)$ which for all x converge to a function f(x) as n goes to infinity, and there is a function $g(x)$ which has a finite integral on the interval we are working in, and that $|f_n(x)| \leq g(x)$ for all n and all x, then we have that $\int \lim_{n\to\infty} f_n = \lim_{n\to\infty} \int f_n$, essentially giving us a condition for when we can swap limits and integrals.

For our purposes, we assume f is integrable, and that f_n is integrable for every n.

Proof. Recall how at some places in level 4 and level 6.1 we talked about the triangle inequality which says that $|a + b|\leq|a| + |b|$, we will use this now to say that

$$|f - f_n| \leq |f| + |-f_n| = |f| + |f_n| \leq 2g$$

Note that f is at most g at all places since it is the pointwise limit of functions that are at most g so it would not make sense for it to exceed g. We also have by definition that $\text{limsup}_{n\to\infty} |f - f_n| = 0$.

Also, in this next step, we will use the fact analagous to the triangle inequality which says that $|\int h(x)| \leq \int |h(x)|$ for any function h(x). For real numbers this is the case because h is always between $-|h|$ and $|h|$, so the integral of h will always be between the integral of $-|h|$ and the integral of $|h|$, so the absolute value of that will be at most the integral of $|h|$. For complex numbers (which I am doing to show that we can generalize this beyond real integrals), we have $\int h = re^{i\theta}$ so $|\int h| = r$ and $\int he^{-i\theta} = r$. Since r is real, we have that $Re(\int he^{-i\theta}) = r$. Since the real part of the integral is the integral of the real part, we have

$$\int Re(he^{-i\theta}) = r \leq \int |he^{-i\theta}| = \int |h|$$

Since $|\int h| = r \leq \int |h|$ this completes the proof.

Now that we have the "integral triangle inequality" we will go back to dominated convergence:

$$\left|\int f - \int f_n\right| = \left|\int (f - f_n)\right| \leq \int |f - f_n|$$

Now, we use the reverse Fatou lemma:

$$\limsup_{n\to\infty} \int |f - f_n| \leq \int \limsup_{n\to\infty} |f - f_n| = \int 0 = 0$$

Therefore, since the lim sup is at most 0 and the terms in $\int |f - f_n|$ are non-negative, both the lim sup and the lim of this sequence must be exactly 0. So, $|\int f - \int f_n| \leq \int |f - f_n| \to 0$ so $|\int f - \int f_n| \to 0$ so $\int f_n$ approaches $\int f$ so the limit and integral interchange is justified.

□

Now, if $f_n(x)$ is defined as n for 0<x<1/n then the integral of this will be 1, but at all points between 0 and 1 f_n will eventually go to 0, so the integral of the limit is not the limit of the integral. It turns out that in this case it turns out we cannot find a function g such that for all n we have $\left|\int_0^1 f_n\right| \leq \left|\int_0^1 g\right| < \infty$. This is a standard textbook counterexample.

3.3 Fubini's theorem

We will only prove a special case of Fubini's theorem.

The theorem says that if I have $\iint f(x, y)\,dxdy$ on a bounded rectangle then I can swap the integrals provided provided f is continuous. Specifically, $\int_{y_0}^{y_1} \left(\int_{x_0}^{x_1} f(x, y)\,dx\right) dy = \int_{x_0}^{x_1} \left(\int_{y_0}^{y_1} f(x, y)\,dy\right) dx$

The claim above is called Fubini's theorem.

A definition: Let f be any bounded function on $[a, b]$, then $\underline{\int_a^b} f(t)dt$ is the supremum over all partitions of $[a, b]$ of the integrals of sums of rectangles that underestimate the function at each partition. This is called the power integral. We similarly define the upper integral as $\overline{\int_a^b} f(t)dt$. f is said to be Riemann integrable if these integrals are equal. By convention we require the interval and f to both be bounded.

We're proving it for riemann integrals, not our extended integrals.

Proof. We will use grid partitions, the X component will be $X_1, X_2, ..., X_n$ and the Y component will be $Y_1, Y_2, ..., Y_m$. We will define the lower integral as

$$\sum_{i,j} |X_i \times Y_j| \inf_{x\in X_i, y\in Y_j} f(x,y)$$
$$\leq \sum_i |X_i| \inf_{x\in X_i, y\in Y_j} \sum_j |Y_j| \inf_{y\in Y_j} f(x,y)$$
$$\leq \sum_i |X_i| \inf_{x\in X_i} \underline{\int_Y} f(x,y)\, dy$$
$$\leq \sum_i |X_i| \sup_{x\in X_i} \overline{\int_Y} f(x,y)\, dy$$
$$\leq \sum_i |X_i|_{x\in X_i} \sup_{x\in X_i, y\in Y_j} \sum_j |Y_j| \sup_{y\in Y_j} f(x,y)$$
$$= \sum_{i,j} |X_i \times Y_j| \sup_{x\in X_i, y\in Y_j} f(x,y)$$

if f is continuous then I can make the second and fifth line differ by an arbitrarily small amount as I refine the partition. I could also swap the roles of X and Y in the proof above. and the middle 2 lines would correspond to the different orders of doing the integrals.

□

Definition. Let g be a function, then we define the Riemann Stieljes integral of $\int f(x)\, dg(x)$ as instead of taking $x_i - x_{i-1}$ we take $g(x_i) - g(x_{i-1})$. When we integrate with respect to a probability measure in Level 6.3, we are essentially setting g to the cumulative distribution function and redefining the "length" for the Lebesgue theorems.

Proposition. If $g$is a cumulative distribution function and f is continuous then f is integrable on any bounded interval.

Proof. The same as the level 4 proof that continuous functions are integrable relied on the fact that showing if our intervals get small enough f will differ by at most εon those intervals. We assumed without loss of generality that the integrals are from 0 to 1 and concluded from there, and the proof is exactly the same since the total change of g is never more than 1 as it is a cumulative probability distribution.

□

Note that for improper integrals, provided the integral of the absolute value is finite, then if we make a big enough rectangle the absolute value of the outside things will go to 0, and the limit will not depend on what sequence of rectangles we take. Therefore for improper integrals we will check for absolute convergence.

Theorem. Let f be a riemann integrable function, then its integral in the above sense exists and equals its extended integral

Proof. **Step 1.** We'll prove if for the case f is non-negative

Quoting what I said earlier, $\underline{\int_a^b} f(t)dt$ is the supremum over all partitions of $[a,b]$ of the integrals of sums of rectangles that underestimate the function at each partition. This means trivially that for each sequence $\varepsilon_n \to 0$ there is a sequence of step functions f_n such that $\int f_n \geq \underline{\int_a^b} f(t)dt - \varepsilon_n$. Then by setting $s_n = \max_{k\leq n} f_k$ this is an increasing sequence as we want.

Step 2. We extend to the general case by adding positive and negative parts.

□

Theorem. The theorems above still work perfectly for integrals in 2 or 3 or higher dimensions

Sketch of proof. Define the base sets as cuboids instead of integrals and put the rest of the theorems with exactly the same proof as in the 1D case.

4 Intuition for complex integration

Also, in my exponentials and logarithms video, I briefly mentioned that if a function has a single-valued antiderivative then the integral along a path of that function does not depend on the path. By "integrate along a path", I mean take the sum of the function times the distance you move by on the path (eg if you move from $1.07 + 3.12i$ to $1.08 + 3.14i$ you add $(0.01 + 0.02i)f(1.07 + 3.12i)$), then take the limit of these sums as these distances go to 0. Intuitively, similar to actual integration, taking the sum like this moves you along the antiderivative, which if it is single valued (unlike log, for example) will not allow you to possibly have path dependence.

Note: We will formalize this in level 8.3, the complex analysis tripos course. For now this tells you why it is true.

5 Uniform continuity

Definition. To say A function is Uniformly continuous means that not only are we able to pick a δ for each x, but a δ that works for all x. An example of a function that is not uniformly continuous is 1/x on the open interval (0,1), since if I pick a certain ε, then no matter how small I pick δ, I can go close enough to 0 that the $|x - a|<\delta$ implies $|f(a) - f(x)|<\varepsilon$ condition is not satisfied, so I cannot pick a δ that works for all x.

Theorem. Suppose that f is a function which is continuous on a closed bounded interval, then it is uniformly continuous

Proof. Since we are assuming our function is continuous on a closed bounded interval, the Bolzano Weierstrass theorem applies by boundedness, and the subsequence in question converges to a limit in the interval by closure of the interval.

We will see similarities to the level 4 proof that continuous functions are integrable.

Assume for a contradiction that our function is continuous but not uniformly continuous on our closed bounded interval. Then there is an $\varepsilon > 0$ such that if I pick $\delta = \frac{1}{n}$ there is some x_n, y_n with $|x_n - y_n| < \frac{1}{n}$ but $|f(x_n) - f(y_n)| > \varepsilon$. We know that x_n and y_n both have convergent subsequences converging to places on our closed interval, and in fact they converge to the same place since the difference betweenx_n and y_n gets arbitrarily small. If the subsequences in question are x_{n_i} and y_{n_i} and their limits are x, then $f(x_{n_i}) \to f(x)$ and $f(y_{n_i}) \to f(x)$ because continuous have functions that whenever the inputs are close the outputs are close, so intuitively you can pass limits through continuous functions like this. But $f(x_{n_i})$ and $f(y_{n_i})$ somehow converge to the same limit but are always $> \varepsilon$, which is a contradiction.

□

6 Calculus results

6.1 Big O and little o notation

Definition. A function $g(x)$ is said to be $O(f(x))$ if when x is sufficiently close to x_0, $|g(x)|$ is bounded by $M|f(x)|$ where M is some fixed positive constant. If $f(x)$ is not 0 in the vicinity of x_0 we can write that $|\frac{g(x)}{f(x)}| < M$ when x is sufficiently close to x_0. We can also say, and this is often done in computer science when analyzing how long an algorithm will take, that a function is a function $g(x)$ is $O(f(x))$ as x goes to infinite if when x is large enough, $|g(x)|$ is bounded by $M|f(x)|$. The value of x_0 in a specific situation is often implied by context. Sometimes people write $g(x) = O(f(x))$

Definition. Little o notation is like big O notation except if $g(x)$ is $o(f(x))$ at x_0 it means that $g(x)$ gets much smaller than $f(x)$ when x is sufficiently close to x_0. Precisely, it means that M can be made arbitrarily small by making x sufficiently close to x_0, or by making x sufficiently large in the case x_0 is infinity.

Note that if $f(x) = o(g(x))$, then we necessarily also have $f(x) = O(g(x))$.

Example. $x \neq O\left(x^2\right)$ as $x \to 0$ since x^2 is much smaller than x. In fact, $x^2 = o\left(x\right)$ as $x \to 0$.

$x = O\left(x^2\right) = o\left(x^2\right)$ as $x \to \infty$ but $x^2 \neq O\left(x\right)$ as $x \to \infty$.

$x = O\left(\sqrt{x}\right)$ as $x \to 0$.

$\sin(2x) = O\left(x\right)$ as $x \to 0$, but $\sin(2x) \neq o\left(x\right)$ as $x \to 0$.

$x \neq O\left(x \sin(x)\right)$ as $x \to \infty$, because although x does not seem to be much larger than $x\sin(x)$, the ratio $\frac{x}{x\sin(x)}$ is unbounded when x gets close to integer multiples of π.

Also, a principle here is the idea that a polynomial is dominated by its leading term. For example

$2x^3 + 4x + 12 = O(x^3)$ as $x \to \infty$ because clearly eventually x^3 exceeds $4x + 12$ so the whole thing is less than $3x^3$ so we can put $M = 3$. Also, big o notation can capture the idea that exponentials beat polynomials (by the way you can prove this as $\frac{a^x}{x^b}$ goes to infinity as x goes to infinity which you can see if you apply l'hopital's rule at least b times), as we can write that $P(x) = o(\exp(x))$ for any polynomial P. Also, $\log\left(x\right) = o(x^{\varepsilon})$ for all positive ε and this can be proven the same way, or by thinking of it intuitively as a sort of inverse of the exponentials beat polynomials idea. Note $\exp(x)$ means e^x.

Another idea is that non-zero constants don't matter here since we're looking at the bigger picture with the growth rate of these functions.

Example. Since $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = f'(x)$, $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} - f'(x) = 0$ so $\frac{f(x+h)-f(x)}{h} - f'\left(x\right) = o(1)$ as x goes to h by definition. Therefore, $f\left(x+h\right) - f(x) - hf'\left(x\right) = o(h)$ by multiplying by h on both sides, so we can write that $f\left(x+h\right) - f\left(x\right) = hf'\left(x\right) + o(h)$, which looks a lot like what I did when I justified differentiating infinite power series.

6.2 Taylor's theorem

Definition. A taylor polynomial of degree n is a taylor series for a function that is n times differentiable but only up to the term in x^n. We write this as $T_n(f)(x)$. This is a polynomial which gives an approximation of f near some point x_0, with error equal to $o((x-x_0)^n)$.

Claim. If f is n times differentiable at x_0 then $f\left(x\right) - T_n\left(f\right)\left(x\right) = o((x-x_0)^n)$ as $x \to x_0$ if x_0 is the point we are doing a taylor series around.

Claim. If f is n+1 times continuously differentiable on the interval $[x_0,\ x]$ then $f\left(x\right) - T_n\left(f\right)\left(x\right) = O((x-x_0)^{n+1})$ as $x \to x_0$ if x_0 is the point we are doing a taylor series around.

Claim. If the conditions for claim 2 hold, then there exists some $t \in (x_0,\ x)$ such that $f\left(x\right) - T_n\left(f\right)\left(x\right) = \frac{(x-x_0)^{n+1}}{(n+1)!} f^{(n+1)}(t)$.

We will prove this.

Lemma. (Extreme value theorem) Any function continuous on a finite closed interval is bounded and actually attains its minimum and maximum.

Proof. Note: It is very important that the interval is closed. On the open interval (0, 1) we can define 1/x which is continuous on that interval, but not on the closed interval [0,1] because the closed interval contains $x = 0$ but the open interval does not. But there's not much you can do to make a continuous unbounded function defined on a finite closed interval – Try it!

We will use the fact that every bounded sequence has a convergent subsequence since that was proven in level 4.

Suppose for a contradiction f is unbounded. Then for all n, we can choose x_n in [a,b] with $|f(x_n)| > n$. Now pick a convergent subsequence x_{n_k} of the sequence formed by each each x_n, and call its limit L, which is in $[a, b]$ since $[a, b]$ is closed so limits of sequences in [a,b] are in [a,b]. By continuity, $f(x_{n_k}) \to f(L)$, but since $f(x_{n_k})$ is unbounded, it follows that this is a contradiction, since $f(L)$ cannot be defined.

Important note: Bolzano weierstrass holds for sequences of points in k dimensions, provided the bounded set the sequence is in is closed as mentioned above – there is a subsequence whose x coordinate converges and a subsequence of that whose y coordinate converges etc, and thus the above theorem holds for functions of more than one variable, as well as the theorem about uniform continuity and about riemann integrals for continuous functions, as the definition of continuity is analagous. This is important since we use these theorems in the context of functions of multiple variables.

We will now prove that the maximum is attained. By boundedness there is a least upper bound M. Therefore there is a sequence of increasing points $M_1, M_2, \ldots$ converging to M that are values of the function. For each of these points pick an input value t_k of f such that $f(t_k) = M_k$. Now t_k has a convergent subsequence converging to t and by continuity $f(t) = M$, meaning the maximum value of a function on a closed interval is attained. It is by closedness that we can gurantee t is in our interval.

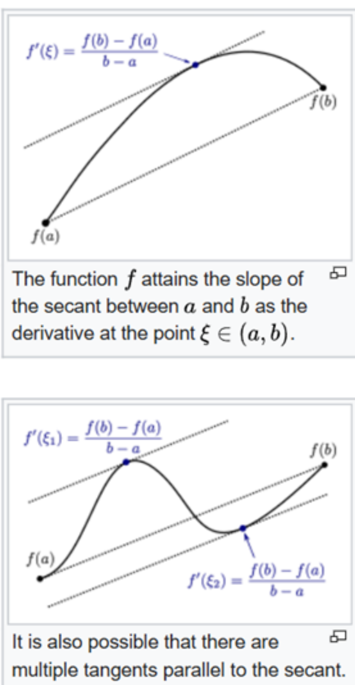
□

Lemma. (Mean value theorem)

If we have a continuous function on [a,b] differentiable on the open interval (a,b) then at some point in (a,b) the derivative is equal to its mean value.

This lemma assumes the values are real, so everything proven here applies to real numbers, but not general complex numbers.

Proof. First of all, a picture (Figure 1, from wikipedia):



The function f attains the slope of the secant between a and b as the derivative at the point $\xi \in (a, b)$.

It is also possible that there are multiple tangents parallel to the secant.

Figure 1

You can see that it's kind of obvious that the function's derivative attains its mean value, since the function is differentiable everywhere by assumption so there are no spikes. This is essentially what the theorem says. However, it is not completely obvious as derivatives can be discontinuous.

To prove this, we want to subtract the line from $(a, f(a))$ to $(b, f(b))$ from y. This line has the slope we want so we want to prove that our new function, which is 0 at a and b, attains a derivative of 0 between a and b. This new function is either constant, in which case the theorem is obvious, or it goes above 0 at some point (If not prove the theorem for minus the function). Since it is continuous on a closed, bounded interval, it is bounded, and has either a maximum or a minimum. At such a point, the derivative exists by our hypothesis, and it can't possibly be something other than 0. Why? Suppose the derivative were positive at a maximum point c, then by definition of limits, $\frac{f(x)-f(c)}{x-c}$ would be positive if we make x close enough to c with $x>c$, implying $f(x)>f(c)$ which is a contradiction.

In particular, we have Rolle's theorem, which says that if $f(a) = f(b) = 0$ and f is differentiable on (a, b) then there is a point $a<x<b$ such that $f'(x) = 0$

□

Lemma. Let f be continuous on $[a, b]$ and n-1 times continuously differentiable, but also n times differentiable on the open interval (a,b), and suppose also that

$$f\left(a\right) = f'\left(a\right) = f''\left(a\right) = \cdots = f^{(n-1)}\left(a\right) = f\left(b\right) = 0$$

then there is some x in (a,b) such that $f^{(n)}\left(x\right) = 0$.

Proof. Induct on n. The $n = 0$ base case is just Rolle's theorem. Suppose we have $k < n$ and $x_k \in (a, b)$ such that $f^{(k)}(x_k) = 0$. Since $f^{(k)}(a) = 0$, we can find $x_{k+1} \in (a, x_k)$ such that $f^{(k+1)}\left(x_{k+1}\right) = 0$ by Rolle's theorem. So the result follows by induction.

□

Example. The function $x^3 - x^2$ has its first derivative and value equal to 0 when $x = 0$, and its value equal to 0 at $x = 1$, so between 0 and 1 it must have a point of inflection. Indeed it does when $x = \frac{1}{3}$.

Proposition. If $f\left(x\right)$ is n times differentiable at $x = x_0$ then

$$f\left(x\right) - f(x_0) - \left(x - x_0\right) f'(x_0) - \frac{\left(x - x_0\right)^2}{2!} f''(x_0) - \cdots - \frac{\left(x - x_0\right)^n}{n!} f^{(n)}(x_0)$$

is $o(\left(x - x_0\right)^n)$ as $x \to x_0$.

Proof. The first n derivatives of the above expression are all 0 when evaluated at x_0 because for k between 0 and n inclusive the $\frac{(x-x_0)^k}{k!} f^{(k)}(x_0)$ term has k'th derivative equal to $f^{(k)}(x_0)$ and all its other derivatives are 0 at x_0 by repeated application of the power rule so therefore it cancels the k'th derivative of the $f\left(x\right)$ term to make it equal to 0.

So, by the definition of o, we need to show that $\frac{f(x)}{(x-x_0)^n} \to 0$ as $x \to x_0$. This is fine as $\left(x - x_0\right)^n$ is not 0 when you make it arbitrarily small so it can go in the denominator.

Now, if we can show that any function whose first n derivatives are all 0 at x_0 is $o(\left(x - x_0\right)^n)$, we will be done. It is clear that a function whose value and first derivative is 0 at x_0 is $o(x - x_0)$ since $\frac{g(x)}{x-x_0} = \frac{g(x)-g(x_0)}{x-x_0} \to g'\left(x_0\right) = 0$, so this theorem is true if n=1. But now suppose that we know that this theorem is true for n=k, then we will prove it is also true for n=k+1, therefore proving this by induction. Let g(x) be such that at x_0 its value and first k+1 derivatives are all 0. By the induction hypothesis, g'(x) has its first k derivatives and value equal to 0 at x_0 so $g'\left(x_0 + h\right) = h^k \varepsilon(h)$ with $\varepsilon\left(h\right) \to 0$ as $h \to 0$, by the definition of little o notation and the induction hypothesis. By the mean value theorem, there exists some ϑ between 0 and 1 such that

$g'\left(x_0 + \vartheta h\right) = \frac{g(x_0+h)-g(x_0)}{h}$, or equivalently $hg'\left(x_0 + \vartheta h\right) = g\left(x_0 + h\right)$, meaning that by the induction hypothesis $g\left(x_0 + h\right) = h\left(h^k \varepsilon\left(\vartheta h\right)\right) = h^{k+1} \varepsilon\left(\vartheta h\right) = o(h^{k+1})$. So induction done and proof complete.

□

Remark:

Just because each remainder is small does not mean that the taylor series converges, since convergence at some x requires that there, the remainder is uniformly small, but it could be that the interval where the remainder is small shrinks and is zero in the limit. There are functions such that the taylor series does not converge in any interval despite the functions being infinitely differentiable, like $\exp(-x^{-2})$ around x=0. However, it turns out that if the function has a complex derivative the taylor series always converges in some interval, see complex analysis in a later level. I'm not proving it because we don't need to use it for this purpose, but it trivializes all of these results.

Lemma. Let f be n+1 times differentiable on [a,b] with it's (n+1)'th derivative continuous. Then we have that

$$f\left(b\right)=f\left(a\right)+\left(b-a\right)f'\left(a\right)+\frac{\left(b-a\right)^{2}}{2!}f^{(2)}\left(a\right)+\cdots+\frac{\left(b-a\right)^{n}}{n!}f^{(n)}\left(a\right)+\int_{a}^{b}\frac{\left(b-t\right)^{n}}{n!}f^{(n+1)}\left(t\right)dt$$

Proof. We will do this by induction on n. When n=0 the theorem says

$$f\left(b\right)=f\left(a\right)+\int_{a}^{b}f'\left(t\right)dt$$

Which is known to be true. Now we will do integration by parts to get

$$\int_{a}^{b}\frac{(b-t)^{n}}{n!}f^{(n+1)}(t)dt=\left[\frac{-(b-t)^{n+1}}{(n+1)!}f^{(n+1)}(t)\right]_{a}^{b}+\int_{a}^{b}\frac{(b-t)^{n+1}}{(n+1)!}f^{(n+2)}(t)dt$$

$$=\frac{(b-a)^{n+1}}{(n+1)!}f^{(n+1)}\left(a\right)+\int_{a}^{b}\frac{(b-t)^{n+1}}{(n+1)!}f^{(n+2)}(t)dt$$

So the lemma follows by induction.

□

Proposition. If $f\left(x\right)$ is n+1 times continuously differentiable in an open interval containing x_0 then

$$f\left(x\right)-f(x_0)-\left(x-x_0\right)f'(x_0)-\frac{\left(x-x_0\right)^{2}}{2!}f''(x_0)-\cdots-\frac{\left(x-x_0\right)^{n}}{n!}f^{(n)}(x_0)-\ldots$$

is $O(\left(x-x_0\right)^{n+1})$ as $x\to x_0$.

Proof. The lemma above immediately implies the result since $f^{(n+1)}$ is continuous on the closed interval [a,b] and thus bounded there, say it is bounded in absolute value by M. Then the integral is at most $\int_a^b \frac{(b-t)^n}{n!}Mdt=M\frac{(b-a)^{n+1}}{n+1!}$ which is $O(\left(x-a\right)^{n+1})$ as required.

□

Proposition. $f\left(x\right)-f\left(x_0\right)-\left(x-x_0\right)f'\left(x_0\right)-\frac{(x-x_0)^2}{2!}f''\left(x_0\right)-\cdots-\frac{(x-x_0)^n}{n!}f^{(n)}\left(x_0\right)=\frac{(x-x_0)^{n+1}}{(n+1)!}f^{(n+1)}\left(t\right)$ for some t(x) between x_0 and x, provided the first n+1 derivatives of f exist and are continuous from x_0 to x. (In fact we don't need the $n+1$'th derivative to be continuous, but merely for it to exist on the open interval (x_0,x) as that is enough to apply Rolle's theorem).

Proof. The n=0 case is just the mean value theorem.

Consider $g\left(x\right):=f\left(x\right)-f\left(x_0\right)-\left(x-x_0\right)f'\left(x_0\right)-\frac{(x-x_0)^2}{2!}f''\left(x_0\right)-\cdots-\frac{(x-x_0)^n}{n!}f^{(n)}\left(x_0\right)$. This has its value and first n derivatives at x_0 equal to 0 by the same reasoning as we gave in one of the earlier proofs

Set $\frac{g(x)}{(x-x_0)^{n+1}}=C$ when x is our constant. Then $g\left(x\right)-C\left(x-x_0\right)^{n+1}$ as a function of x (so x is no longer constant) is 0 at x from how we defined C and has its first n derivatives and value equal to 0 at x_0 meaning it satisfies the conditions for generalized rolle's theorem. Therefore, there exists a point y between x_0 and x such that the n+1'th derivative of $g\left(y\right)-C\left(y-x_0\right)^{n+1}$ is 0. But the derivative of the second term is just C(n+1)! by the power rule, thus the n+1'th derivative of g at y is also C(n+1)!, thus $C=\frac{g^{(n+1)}(y)}{(n+1)!}$. Therefore, since we have that $g\left(x\right)-C\left(x-x_0\right)^{n+1}$ is 0 at x from earlier, we thus have that

$$g\left(x\right)=C\left(x-x_0\right)^{n+1}=\frac{g^{(n+1)}(y)}{(n+1)!}\left(x-x_0\right)^{n+1}$$

Which is exactly what we wanted to prove.

□

Example. There exists some point between 0 and 1 such that $\frac{e^x}{2} = e-2$, because if $x = 1$ and x_0 =0, then $e^x - x - 1 = \frac{x^2}{2}e^y$ for some y between 0 and 1, but x=1 so this simplifies to $\frac{e^y}{2} = e - 2$.

6.3 L'hopital's rule

Lemma. (Cauchy mean value theorem) Let f, g be functions taking real values continuous on [a,b] and differentiable (a,b). Then there exists a c in (a,b) such that $g'(c)(f(b) - f(a)) = f'(c)(g(b) - g(a))$

Proof. Try the function $h(x) = [g(x) - g(a)][f(b) - f(a)] - [g(b) - g(a)][f(x) - f(a)]$.

Now $h'(x) = g'(x)(f(b) - f(a)) - f'(x)(g(b) - g(a))$

This is 0 at some point because $h(a) = h(b)$ and Rolle's theorem is true.

$\square$

Proof of L'hopital's rule

Suppose that the following hold:

1. $f(a) = g(a) = 0$
2. $g(x)$ is not 0 if x is close to c
3. $\lim_{x \to c} \frac{f'(x)}{g'(x)}$ exists and is equal to l.
4. f and g are differentiable near a

By the Cauchy mean value theorem $g'(c)(f(b)) = f'(c)(g(b))$ for some c between a and b, for any b>a. Rearranging gives that

$$\frac{f(b)}{g(b)} = \frac{f'(c)}{g'(c)}$$

I emphasise that c depends on b but for each fixed b it exists and is between a and b. Therefore, as b approaches a, so does c.

Thus, $\frac{f(b)}{g(b)} \to l$ as $b \to a$.

7 Uniform convergence

We will define uniform convergence.

We know about how pointwise convergence works – a sequence of functions converges to a function if it converges at every value. We can say "For each point, there is an $\varepsilon > 0$ such that for all x, our sequence of functions $f(x)$ is eventually, after some n, within ε of the limit function". This is different from uniform convergence in the sense that uniform convergence requires n not to depend on x. We need the function to eventually get arbitrarily close at all points at once, not just at each point. An example of a sequence of functions that converges pointwise but fails this is the following:

$$f_n(x) = \begin{cases} 0 < x < \frac{1}{n} : 1 - nx \\ Otherwise : 0 \end{cases}$$

This looks like the functions in these images:

Figure 2 shows examples of this for n=1, 2, 10.

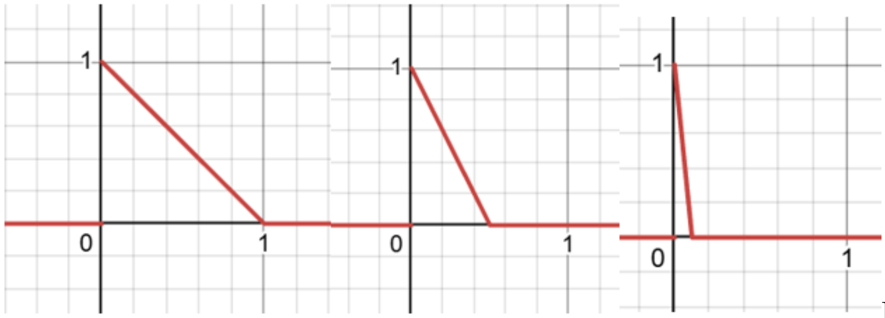


Figure 2

The point is: For any point you choose, say k, after $> \frac{1}{k}$ steps, we will get to 0. So we converge pointwise to 0. It seems like we can pick x=0 and it will converge to 1 but we defined f(0) to be 0. We do not converge uniformly to 0 as we never satisfy that all points get arbitrarily close to 0.

It is useful because, for example, if we want to swap a limit with an integral we know that the total error will eventually get small, rather than just the error at each point.